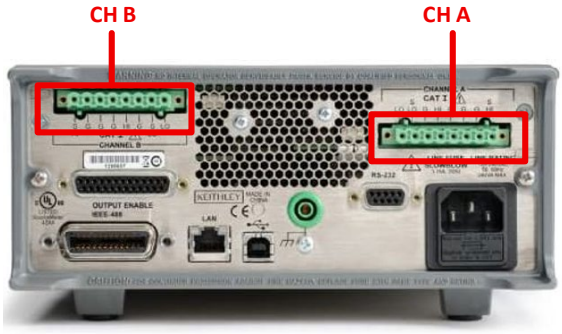
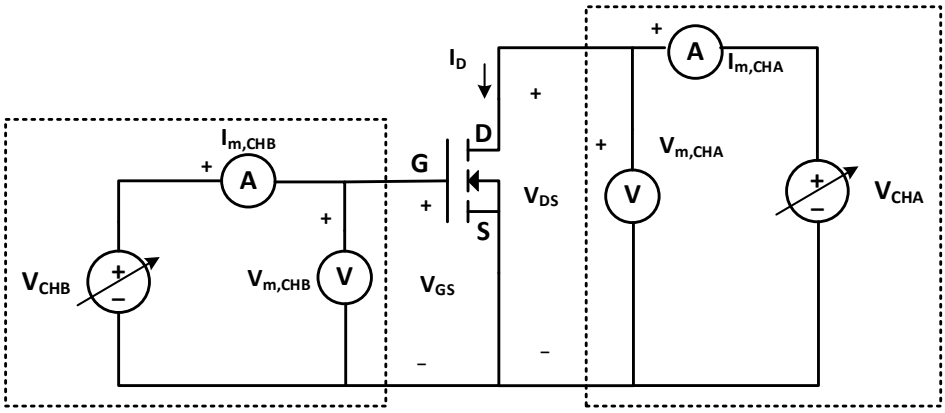
ECE321 Lab 2: Transistor Characterization with Curve Tracer F26v1

Discussion: In this lab, we will use the Keithley 2604B Source Meter to collect data to characterize our ZVN2106 NMOS transistor and find estimates of our four desired model parameters: V_t, K_n, λ, and r_o (the transistor output resistance). As shown below, the 2604B has two channels and we will need them both to control the terminal voltages V_{GS} and V_{DS} while measuring the transistor's I_D. We will use the *Kickstart* software to configure the channel settings and design our intended experiment (you should have had a brief introduction to it in Practical Exercise #4).

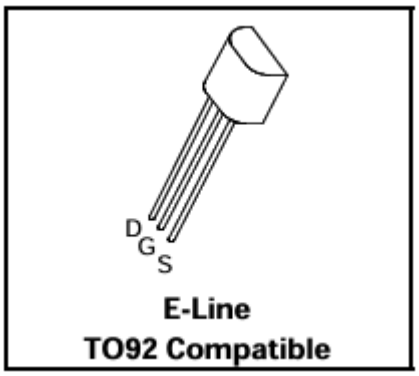


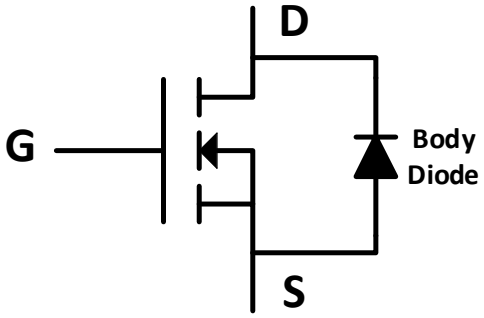


Hardware Setup: The basic lab setup is captured by the figure below. What data is captured and how voltages are varied are controlled by the *Kickstart* software.

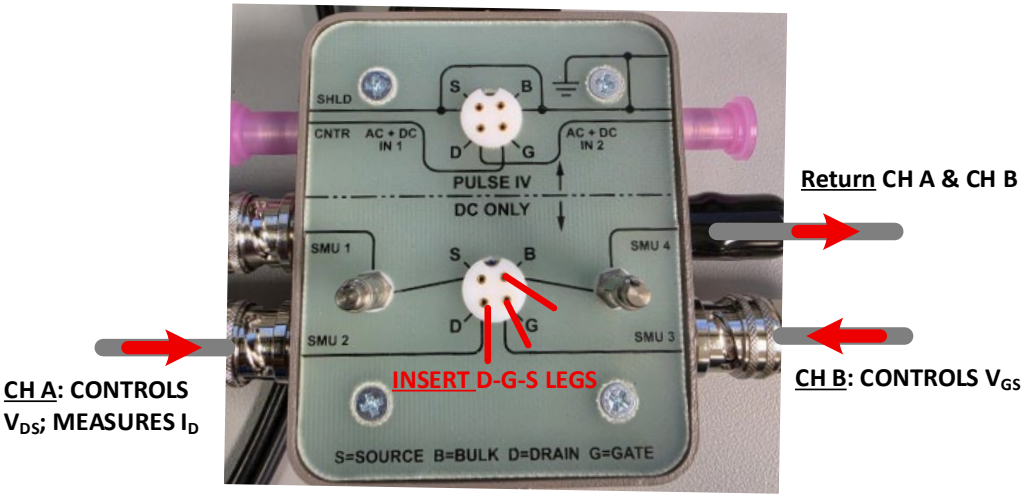


We can realize this setup using the Keithley 8101-PIV test fixture. The test fixture is configured by inserting the ZVN2106 NMOS in the D-G-B terminals as shown on the next page. Note, the transistor pinout is given as follows. *Verify by finding the body diode using the DMM diode function as shown in class.*





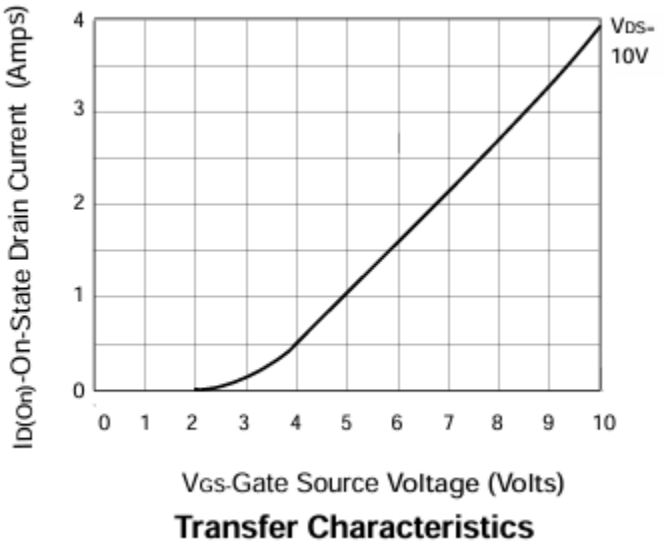
Since the Body and Source are connected internal to the transistor package, the B and S terminals are the same. This setup should remain the same for the three parts of the lab.



PART A: Transfer Characteristic (V_t determination)

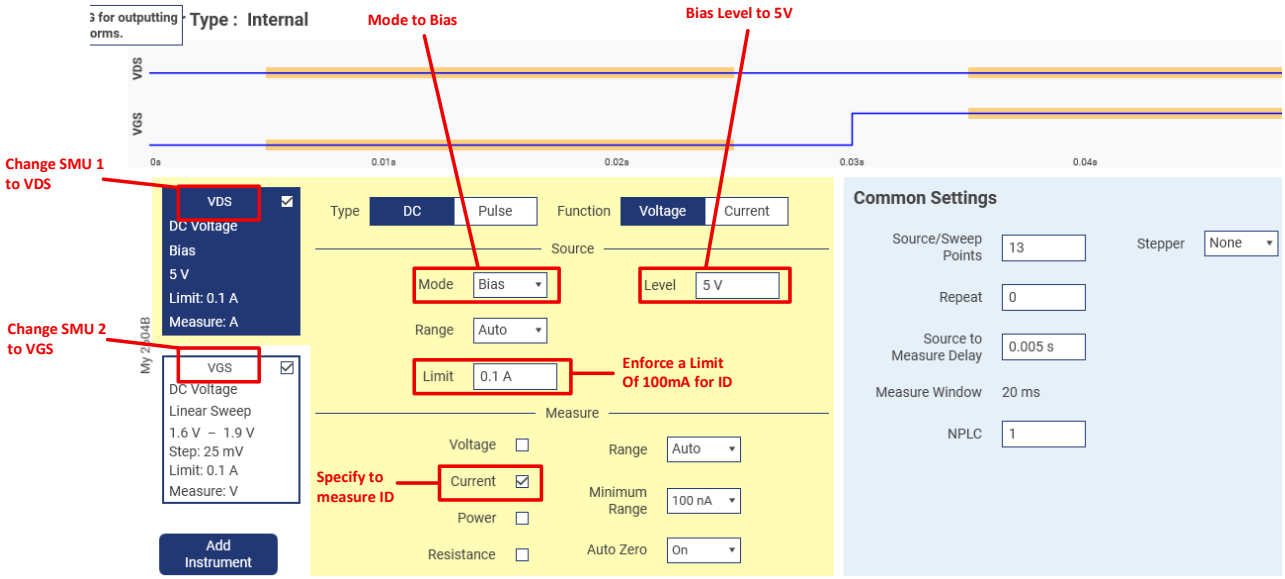
Discussion: We will hold V_{DS} constant (at a value that will be reasonable for future amplifier designs), sweep V_{GS} over values that we might anticipate the threshold voltage, and create the V_{GS} vs I_D curve. The gate voltage where our drain current is sufficiently small is what we will identify as V_t. Given that our amplifier designs will likely focus around drain currents of 1mA to 20mA, *we will pick 50μA as our sufficiently small point*.

The transfer characteristic from the ZVN2106 datasheet is as follows

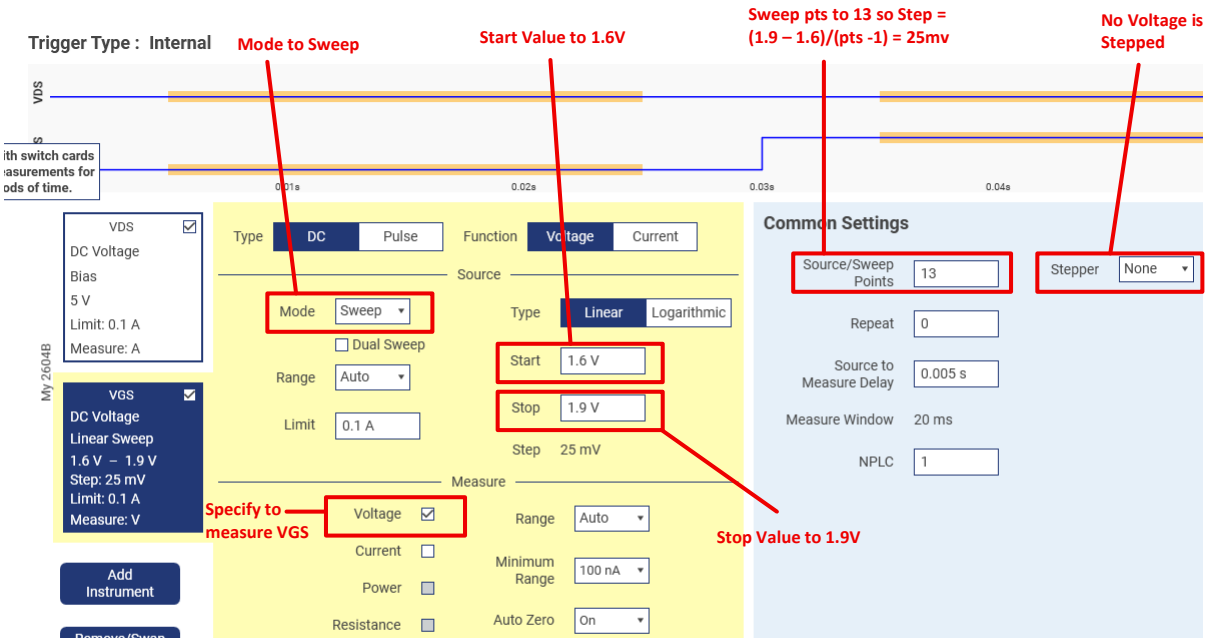


Note, here V_{DS} is held at 10V and we sweep V_{GS} from 0 to 10V. We are anticipating V_t to be in the range of 1.6V to 1.9V, so we will focus our sweep closer to the "action" and thus anticipate much smaller currents as shown above. *We will also hold V_{DS} at a "bias" value of 5V* because that will be a good point to consider in our future amplifier designs.

Let's review how to configure the *Kickstart* software. Open *Kickstart*, choose *My2604B*, then select *IV Characterizer*. This should open up the settings panel (below). Change the labels on SMU1 and SMU2 and configure the VDS settings as shown below.



Now let's move to the second channel and configure the VGS settings



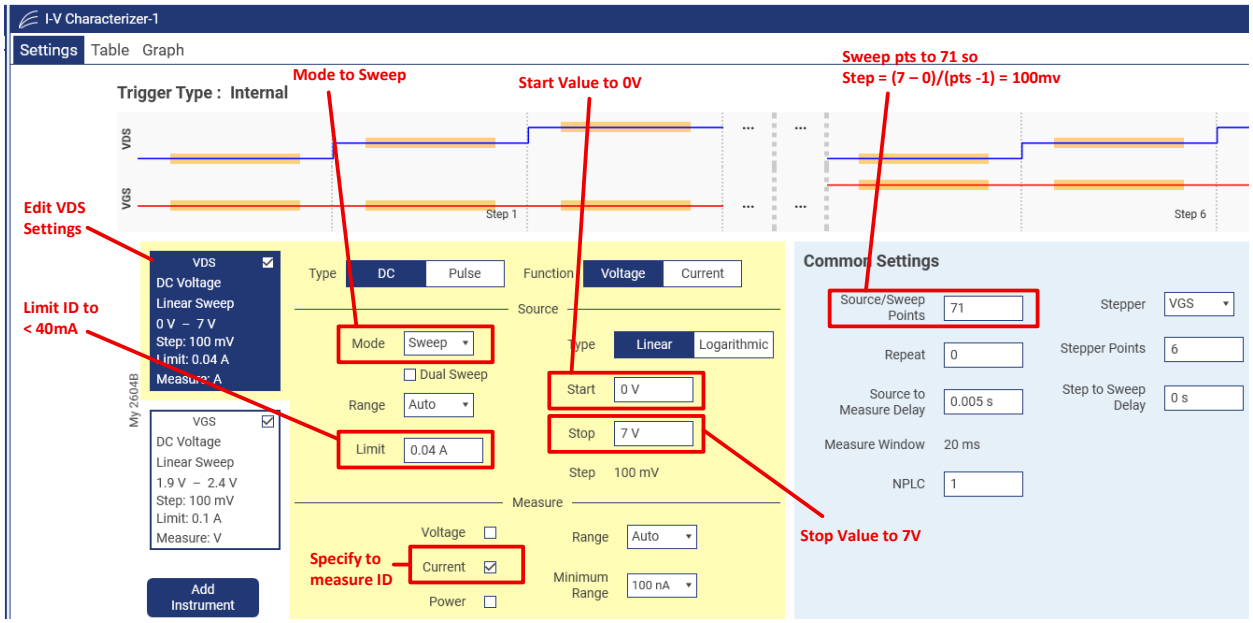
Click Run. Toggle from Settings to Graph and Table. Provided this range of V_{GS} values is adequate (meaning we see a drain current of around 50μA), then export the Transfer Characteristic and Data Table and record the specific value of V_{GS} which will be your threshold voltage. *If your threshold voltage is not in this range, then you must modify "Start" and "Stop" and possibly increase "Sweep Points" to get a data point close to 50μA.*

******* **Report your value to the instructor:** _____

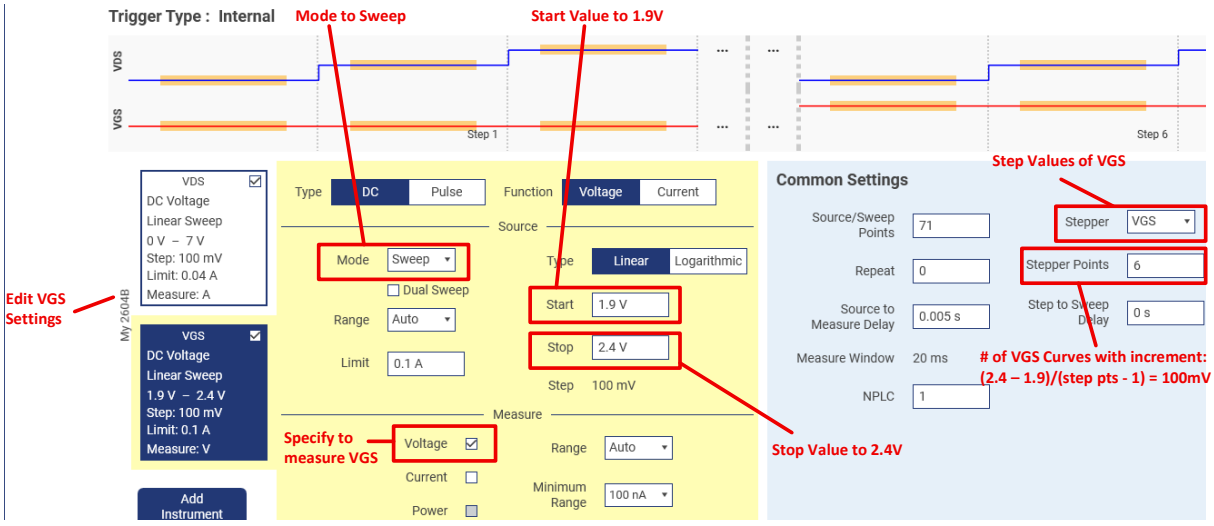
PART B: MOSFET Family of Curves

Next, we will generate our family of curves (I_D vs V_{DS} for specific values of V_{GS}). So, we will be sweeping both V_{DS} and V_{GS}, but we will now want to use the *Stepper* capability so we fix V_{GS} at specific values, then proceed to sweep V_{DS}. Thus, we will need to change our V_{DS} setting mode to sweep and we will choose the *range of 0V to 7V*. If we go much higher, we risk drawing higher I_D values, heating up the device, and seeing weird data. For V_{GS}, we want to create 6 curves all above V_t. The *Kickstart* settings are as shown next.

VDS Channel Settings (note, we are applying a hard limit of 40mA on ID)



VGS Channel Settings (*note, our range of 1.9V to 2.4V depends on Vt. You should adjust it as required based on your PART A findings*)

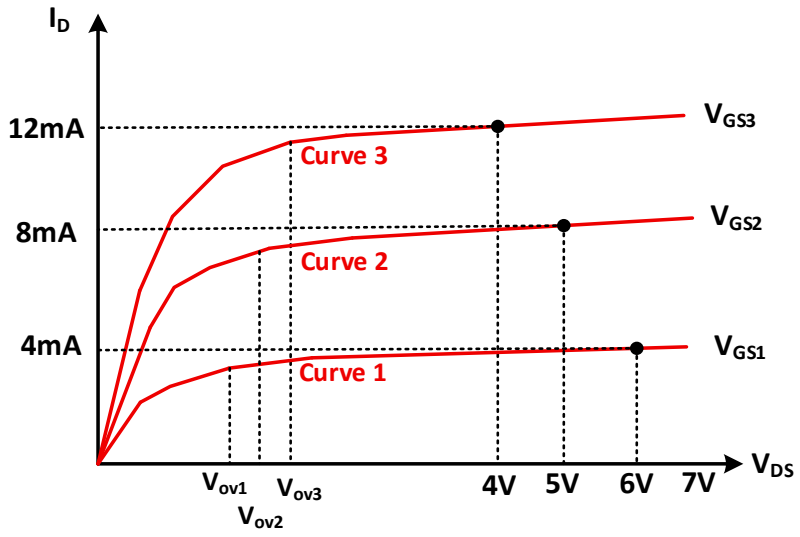


Hit *Run*. Switch to Graph and Table to view your results. Does your curve look as expected? If content, export your Graph and Table (excel)

******* **Have the instructor confirm your results:** _____

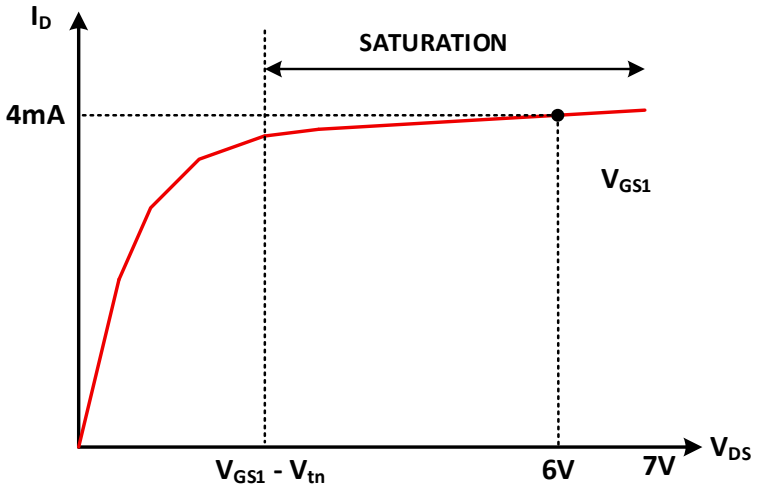
PART C: Specific Data Points of Interest

Now we want to investigate specific ID vs VDS curves to establish values of l, Kn, and ro. We are going to focus in on finding three specific curves that yield saturation currents that we may use in our future amplifier design: 4mA, 8mA, and 12mA. Note, we will want to individually collect these curves

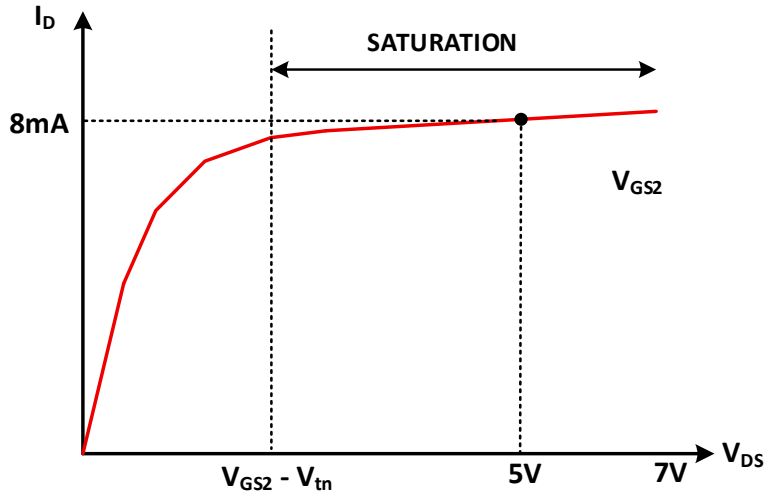


We will start by creating the 4mA curve.

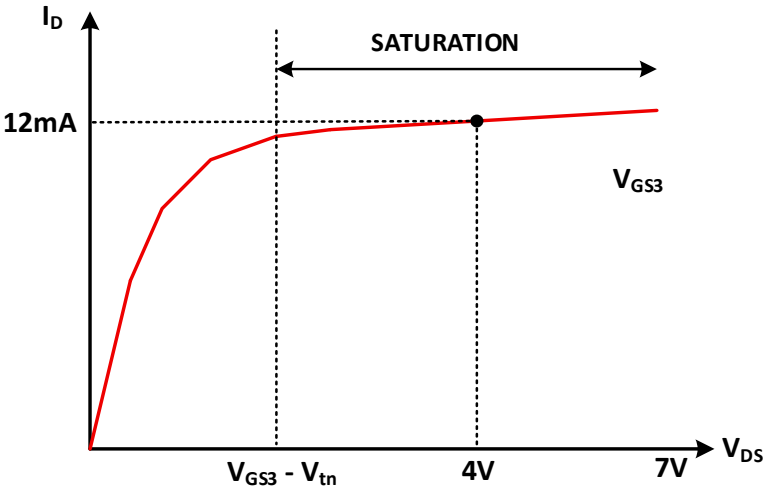
- Change V_{GS} mode from *Sweep to Bias*
- Choose a value for V_{GS} (above V_t but below say 2.4V where the I_D currents might start getting excessive)
- Keep V_{DS} mode on Sweep, Current Limit at 0.04A, and 71 sweep points
- Hit run, view your data. Iterate until your curve *approximately* goes through about V_{DS} = 6V and I_D = 4mA. At that V_{GS} point, export your Graph and Table (excel)



Next, repeat that process to create the following 8mA curve, once again adjust V_{GS} until you *approximately* go through V_{DS} = 5V and I_D = 8mA. At that point, export your Graph and Table (excel).



Finally, repeat one final time to create the following 12mA curve, once again adjust V_{GS} until you *approximately* go through V_{DS} = 4V and I_D = 12mA. At that point, export your Graph and Table (excel).



********* Have the instructor confirm your results: _____**

PART D: Analysis

1. For each of the three tables you exported from PART C, strip off the points that are comfortably in saturation. *To ensure we are not pulling points close to the triode boundary choose values of $V_{DS} > 2^* (V_{GS} – V_t)$ and beyond*
2. Export this excel data into Matlab
3. Use the *polyfit* function to fit a line through your data, identifying the slope m and y-intercept b
4. For each of the three lines, compute the transistor output resistance r_o, the Early Voltage V_A, and the channel length modulation parameter λ. Find the average of the three λ's.
5. Finally, let's compute the MOSFET transconductance K_n for the three data points (V_{DS}, I_D): (6V, 4mA), (5V, 8mA), and (4V, 12mA). DO NOT ignore channel length modulation. Find the average of these three values of K_n.
6. Finally[2], plug your values of V_t, K_n, and λ into the Multisim *Virtual Diode* component and create a family of curves similar to what you produced in PART B. Assess how well your Multisim model captures your data.

Hand In:

- **PART A**: Plot of the transfer characteristic and your reported value of $V_{GS} = V_t$ at 50μA. Highlight on your Table of data where you established V_t.
- **PART B**: Plot of Family of Curves and your Table of output data
- **PART C**: Plots of your 4mA, 8mA, and 12mA V_{DS} vs I_D curves.
- **PART D**: Your results and any calculations for questions 3, 4, and 5. Your simulation circuit diagram and the associated Family of Curves. Your assessment of the Multisim output.
- *NO formal lab report or lab notebook submission* as the procedure is captured in this document.